João Barão

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LinkedIn | GitHub

RESEARCH INTERESTS

My research interests lie in understanding what reinforcement learning (RL) teaches LLMs during post-training. In particular, I am interested in (i) open-ended and long-horizon environments, where sparse rewards make training difficult; (ii) the capabilities that RL with verifiable rewards (RLVR) produces, and whether they transfer to other settings; and (iii) reward hacking, measuring when a policy starts optimizing the judge instead of the task. My main research question is: *how can we extend RL post-training beyond verifiable tasks, to open-ended and long-horizon problems?* One of my long-term goals is to develop systems capable of open-ended scientific discovery.

EDUCATION

University College London (UCL) Sept. 2025 – Sept. 2026
MSc Data Science – Graduated with Distinction. London, UK

University of Porto – FCUP / FEUP Sept. 2022 – Jul. 2025
BSc in Artificial Intelligence and Data Science Porto, Portugal

- **Best Student** of the BSc in Artificial Intelligence and Data Science; achieved the **highest GPA in the programme's history**.
- Co-founded **JuniorAI**, Portugal's first AI Junior Enterprise (Sept. 2024 – Jul. 2025).

RESEARCH & TECHNICAL EXPERIENCE

INESC-ID 2026 – Present
Research Assistant – CMU Portugal Exploratory Research Project Lisbon, Portugal

- Working on QSHIELD, a joint INESC-ID / CMU project funded by FCT, with Prof. José Fragoso Santos (INESC-ID) and Prof. Limin Jia (CMU).

Miles AI 2025
Co-founder & CEO London, UK

- Co-founded a startup, backed by Sterling Road VC, building LLM and retrieval systems that draft replies to customer emails.
- **Finalist** in Imperial College London's IdeaForge startup incubator 2026 (**top 5 of 100+ teams**).

University of Porto – FCUP 2024 – 2025
Invited Teaching Assistant – Numerical Methods Porto, Portugal

- Supported undergraduate lab sessions, grading, and explanation of numerical algorithms and implementation exercises.

NOS SGPS 2024
Data Scientist Intern Porto, Portugal

- Developed and evaluated ML models for uncertainty quantification.

SELECTED RESEARCH & TECHNICAL PROJECTS

Hybrid LLM–Human Relevance Judgments for Evaluating Top Retrieval Systems 2026

- Submitted to **ECIR 2027**. MSc thesis at UCL, supervised by Prof. Ingemar Cox.

A Fine-Tuning Approach to Endangered Language Preservation and Word-Level Translation 2026

- Fine-tuned LLaMA 3.1 8B and Gemma 3 4B with QLoRA for word-level translation of Chakma, an endangered language.

Vota AI | *RAG, NLP, LLMs* Featured on National TV May 2025

- Co-developed a free RAG chatbot on party programmes for the Portuguese general elections, **used by 50,000+ people**.

SELECTED AWARDS

Academic Distinction Scholarship "Impulso Jovens STEAM";
Finalist, Portuguese Mathematical Olympiad